FATMA GÜMÜŞ, PhD

Assistant Professor | Department of Computer Engineering, Turkish Air Force Academy (Hava Harp Okulu), National Defence University (Milli Savunma Üniversitesi), Istanbul, Türkiye

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RESEARCH PROFILE

Computer engineering researcher working across trustworthy and robust machine learning, adversarial AI, speech and audio processing, computer vision, and learning from imperfect data. Research includes black-box attacks on language and speech systems, synthetic speech detection, structured bounding-box annotation noise, and reliable visual recognition. Interested in a six-month visiting research collaboration in 2027.

RESEARCH INTERESTS

Robust and adversarial machine learning • Speech/audio AI and synthetic speech detection • Computer vision and object detection • Annotation noise and reliable learning • Distribution shift and uncertainty-aware AI • Adaptive learning

ACADEMIC APPOINTMENTS

2022–present | Assistant Professor, Computer Engineering, Turkish Air Force Academy (Hava Harp Okulu), National Defence University (Milli Savunma Üniversitesi)

2019–2022 | Research Assistant, Computer Engineering, Turkish Air Force Academy (Hava Harp Okulu), National Defence University (Milli Savunma Üniversitesi)

EDUCATION

PhD, Computer Engineering — Yıldız Technical University (Yıldız Teknik Üniversitesi), 2021. Dissertation: “Yapay öğrenme algoritmalarını kandırmak” (Deceiving Machine Learning Algorithms). Advisor: Mehmet Fatih Amasyalı.

MSc, Computer Engineering — Istanbul University (İstanbul Üniversitesi), 2016. Thesis: “Yazılım tanımlı ağlarda makine öğrenme algoritmaları ile tıkanıklık kontrolü” (Congestion Control Using Machine Learning Algorithms in Software-Defined Networks).

BSc, Computer Engineering — Istanbul University (İstanbul Üniversitesi), 2013.

JOURNAL PUBLICATIONS

Gümüş, F. (2026). “NoisyBox: a dataset for analyzing structured spatial annotation noise in bounding boxes.” PeerJ Computer Science. Accepted; scheduled for publication on 22 October 2026. [DOI pending publication.]

Aksu Demir, İ. E., & Gümüş, F. (2026). “Source-Rate Representations and Attention Autoencoders for Leakage-Safe Zero-Day Sybil Detection in UAV Networks.” Electronics, 15(17), 3966. https://doi.org/10.3390/electronics15173966

Gümüş, F., & Amasyalı, M. F. (2025). “Robustness of emotion recognition in dialogue systems: A study on third-party API integrations and black-box attacks.” Speech Communication, 175. https://doi.org/10.1016/j.specom.2025.103316

Gümüş, F., & Amasyalı, M. F. (2022). “Exploiting Natural Language Services: A Polarity Based Black-box Attack.” Frontiers of Computer Science, 16(5).

Gümüş, F., & Yıltaş Kaplan, D. (2020). “Congestion Prediction System With Artificial Neural Networks.” International Journal of Interdisciplinary Telecommunications and Networking, 12(3), 28–43. https://doi.org/10.4018/IJITN.2020070103

Karaman, F., & Gümüş, F. (2025). “Exploring the Impact of Model Capacity and Parameter Tuning on 3D Semantic Segmentation.” Turkish Journal of Science and Technology, 20. https://doi.org/10.55525/tjst.1637713

Gümüş, F., & Saylam, R. (2025). “Optimizing EEG-Based Cognitive State Classification with Feature-CNN in Aviation: A Comparative Analysis.” Journal of Aeronautics Astronautics and Aviation, 18(1), 145–161.

Gümüş, F., & Eyüpoğlu, C. (2024). “An Image Processing-Based Approach for Reading Needle-Type Instruments on Aircraft.” ELECTRICA, 24(2), 425–435. https://doi.org/10.5152/electrica.2024.23190

Gümüş, F., Ata, O., & Balık, H. H. (2018). “Davranışsal Biyometrinin 5 Yılı: Kimlik Doğrulama ve Anomali Tespit Uygulamaları” (Five Years of Behavioral Biometrics: Authentication and Anomaly Detection Applications). Fırat Üniversitesi Mühendislik Bilimleri Dergisi (Fırat University Journal of Engineering Sciences), 30(1), 345–364.

SELECTED CONFERENCE PAPERS

Karaman, F., Gümüş, F., & Eyüpoğlu, C. (2026). “Cross-Lingual Detection of Synthetic Speech Using Acoustic and Deep Features.” International Informatics and Software Engineering Conference (IISEC). https://doi.org/10.1109/IISEC69317.2026.11418501

Gümüş, F. (2024). “Reliability Challenges in Crowdsourced Bounding-Box Data for Computer Vision.” ICAIAME 2024. https://doi.org/10.1007/978-3-031-92552-8_4

Çakı, B., & Gümüş, F. (2026). “Multimodal Remote Sensing Image Segmentation with Domain Adaptation: A Comparative Review.” RAST 2026. https://doi.org/10.1109/RAST69551.2026.11672343

Yiğit, G., Gümüş, F., & Yeğin, M. N. (2026). “The Effects of Prompting and Fine-Tuning in Medical Text Simplification: Evaluation of Various Models.” 10th International Artificial Intelligence and Data Processing Symposium.

Gümüş, F., & Eyüpoğlu, C. (2023). “A Survey of Prominent Image Processing Research on Needle-type Instrument Reading on Aircraft.” RAST 2023. https://doi.org/10.1109/RAST57548.2023.10197881

TEACHING & SUPERVISION

Courses taught (selected): Artificial Intelligence, Machine Learning, Image Processing, Digital Audio and Speech Processing, Natural Language Processing, Theory of Computation, and Capstone Projects; undergraduate, master’s, and doctoral levels.

Master’s thesis supervision (2026): synthetic speech detection using spectrogram representations and meta-learning (completed); speaker representations and diarization (ongoing); UAV cyberattack detection (completed).

INDUSTRY EXPERIENCE

2015–2018 | R&D Software Engineer, Karaş Yazılım (Karaş Software Development). Machine-learning-based speaker verification and biometric speech analysis; TÜBİTAK (Scientific and Technological Research Council of Türkiye) R&D project proposal and progress management.